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PROFESSIONAL SUMMARY

AI Engineer specializing in Generative AI, Conversational AI, and LLMs, with hands-on experience building production-grade RAG pipelines, agentic systems, and multi-agent applications. Proven track record of taking AI products from research to production, delivering enterprise solutions that automate complex workflows and serve real users at scale. Strong foundation in Data Science and data-driven experimentation.

WORK EXPERIENCE

AI Engineer

AiTech

January 2025 – Present

Giza, Egypt

- Built and deployed a multi-tenant **enterprise AI customer service chatbot platform** serving multiple organizations, reducing response times from 24 hours to real-time with **90% query resolution**.
- Engineered a production-grade **hybrid RAG pipeline** with binary quantization, BM25 retrieval, Reciprocal Rank Fusion, and Cross-Encoder reranking, improving retrieval precision while reducing memory consumption, extended the knowledge base to support image documents by integrating Vision-Language Models and PaddleOCR with confidence-threshold routing for accurate multimodal retrieval.
- Designed a dynamic **multi-model orchestration layer** routing across self-hosted vLLM and external providers (OpenAI, Gemini, Mistral), including load balancing, health monitoring, and automatic failover.
- Implemented AI safety features: hallucination detection, prompt injection prevention, semantic caching, confidence-scored self-reflection loops, guardrails, and **Human-in-the-Loop (HITL)** escalation.
- Built an **LLM-driven load-testing framework** simulating 100+ context-aware personas in multi-turn conversations for large-scale behavioral validation and stress testing.
- Developed token monitoring and quota management services to track LLM consumption, enforce SaaS client usage limits, and optimize operational AI costs.
- Scraped and analyzed **4,500+ monthly posts** from Facebook, Reddit, and Twitter for a brand monitoring platform, achieving **85% topic classification accuracy** for sentiment, themes, and competitor signals, exposed a RAG-powered chatbot for stakeholder queries with automated visual dashboards.
- Achieved **94% speaker diarization accuracy** processing **80+ hours of recordings per week** at 25 concurrent requests, built a multi-agent pipeline (CrewAI + vLLM) for automated transcription and structured meeting summarization.
- Backed the AI assistant with FastAPI, Celery, and Redis for async task management, integrated a Telegram Bot as a delivery channel for real-time user interactions.

AI Engineer Instructor (Part-time)

IctHub

July 2026 – Present

Cairo, Egypt

- Deliver hands-on training in Machine Learning, Deep Learning, and Generative AI to aspiring AI engineers, combining strong theoretical foundations with practical implementation through real-world projects.

Military Service | Ministry of Interior – Egypt

November 2023 – December 2024

PROJECTS

DocuMind – AI Document Intelligence Platform | [GitHub](#)

GenAI, RAG, FastAPI, LangChain, Docker

- Built a multimodal AI platform supporting PDFs, Office files, images, audio, video, websites, and YouTube using LlamaParse, PaddleOCR, Mistral Vision, and cloud/local LLMs.
- Implemented hybrid RAG with semantic embeddings, BM25 retrieval, citation-aware responses, intelligent chunking, and multilingual prompting to minimize hallucinations.
- Designed a scalable backend with FastAPI, Celery, Redis, MongoDB, and ChromaDB, integrated real-time voice via Vapi/Twilio and Gemini 2.5 Flash TTS for multilingual speech.

MedAgent – Intelligent Medical Research Assistant | [GitHub](#)

LangGraph, FastAPI, Streamlit, MongoDB

- Orchestrated a multi-agent medical research workflow using LangGraph with live web search (Tavily) and PubMed retrieval, autonomous planning, and task decomposition.
- Implemented self-reflection loops and real-time evidence verification to reduce hallucinations and improve clinical insight reliability.
- Delivered structured research reports with source attribution, enabling evidence-based Q&A for medical professionals.

Alzheimer Disease Prediction System | [GitHub](#)

Scikit-learn, XGBoost, Random Forest, SVM

- Trained an ensemble classifier (XGBoost, RF, SVM) on **5,000+** patient records achieving **92.3% accuracy** and **AUC-ROC of 0.95** across 15 cognitive and biomarker features.
- Applied SMOTE class balancing, GridSearchCV hyperparameter tuning, and SHAP interpretability analysis for clinical explainability.

EDUCATION

Bachelor of Computer Science	2019 – 2023
<i>Modern Academy for Engineering and Technology</i>	<i>Cairo, Egypt</i>
• Graduation Project (Ranked 1st): Dermatection — fine-tuned ResNet50 on 10,000+ dermatology images achieving 94% accuracy across 7 disease categories, deployed via Flask REST API with TensorFlow Serving and a rule-based patient guidance chatbot.	

CERTIFICATIONS

<u>Oracle Cloud Infrastructure Generative AI Professional</u>	<i>Oracle Cloud, 2025</i>
<u>Oracle Cloud Infrastructure AI Foundations Associate</u>	<i>Oracle Cloud, 2025</i>
<u>LLM Engineering: Master AI & Large Language Models</u>	<i>Udemy, 2025</i>
<u>Generative AI with LangChain & Hugging Face</u>	<i>Udemy, 2025</i>

SKILLS

LLM Frameworks: LangChain, LangGraph, CrewAI, Hugging Face Transformers, vLLM, Ollama
Model Techniques: LoRA, QLoRA, Fine-Tuning, Transfer Learning, Prompt Engineering, Human-in-the-Loop
Vector & Search: Qdrant, ChromaDB, FAISS, BM25, Cross-Encoder Re-ranking, Semantic Caching
Speech & Vision: Whisper, PyAnnote, YOLOv8, PaddleOCR, Vision-Language Models, OpenCV
Backend: FastAPI, Celery, Redis, RabbitMQ, WebSockets, WebRTC, Streamlit, Telegram Bot API, Twilio
Automation: n8n, Typebot
Observability: LangSmith, LangFuse, Distributed Logging, A/B Testing
ML & Data: Scikit-learn, PyTorch, TensorFlow, Pandas, PySpark, Databricks
Databases: MongoDB, PostgreSQL
Cloud & DevOps: Docker, Git, Azure AI Services, Oracle Cloud Infrastructure (OCI)
Languages: Arabic (Native), English (Professional Working Proficiency)